HyperProve

Answer-Guided Hypergraph Expansion for Multi-Hop Question Answering

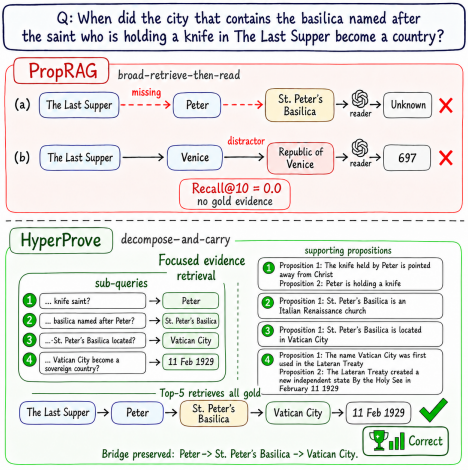
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EMNLP 2026 Main Conference · Budapest, Hungary



Multi-hop evidence hides behind bridges



4-hop chain: *Last Supper* → *Peter* → *St. Peter's Basilica* → *Vatican City* → 1929

- Only the **first** fact resembles the question
- Later facts become relevant *after* a bridge (*Peter*) is resolved
- Fixed-query retrieval (PropRAG): misses *Peter*, drifts to the *Venice* distractor
  ⇒ Recall@10 $= 0$
- HyperProve: decompose + carry the bridge
  ⇒ all gold evidence in top-5

> Relevance **changes after every hop**
> ⇒ retrieval needs an explicit **state**.

Where existing RAG falls short

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- **Dense:** Contriever, BGE-M3
  ⇒ Misses facts unlike the original question.
- **Structured:** HippoRAG2, PropRAG, HyperGraphRAG, HGRAG
  ⇒ Fixed-query expansion can follow distractor paths.
- **Iterative:** IRCoT, HopRAG, ChainRAG
  ⇒ Supporting evidence can be lost between hops.

> We combine **graph structure** with **agentic reasoning** for **more query-aware retrieval**.

Key idea: carry evidence state, not just text

Text state
IRCoT, ChainRAG

hop t answer a_t → (text only) → rewrite q'_{t+1} with the string a_t → dense search over the **whole corpus**

⇒ *bridge evidence is dropped; next hop restarts from surface similarity*

Structural state
HyperProve

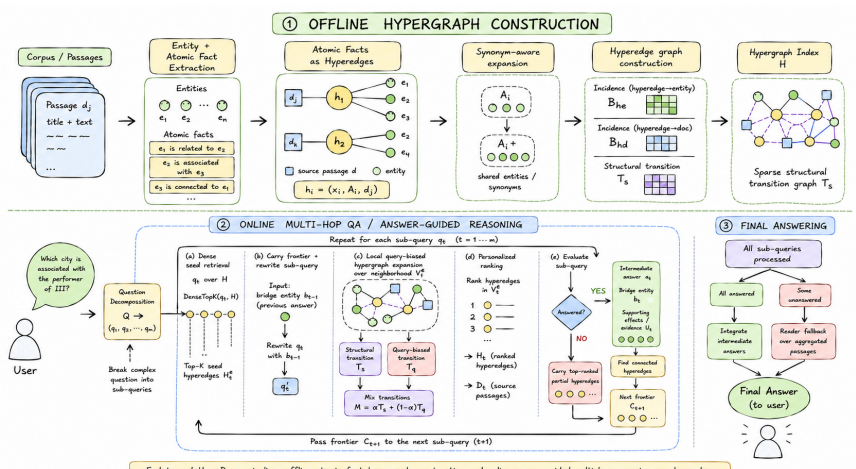
hop t answer a_t → ($a_t + \mathcal{U}_t$) → frontier $\mathcal{C}_{t+1}$: facts around b_t & $\mathcal{U}_t$ → **local walk** from $\mathcal{C}_{t+1}$ biased by q'_{t+1}

⇒ *provenance kept; the next retrieval distribution itself changes*

$$\mathcal{C}_{t+1} = \mathrm{FindHyperedges}\Big(\underbrace{\{b_t\}}_{\text{bridge entity}} \cup \underbrace{\mathrm{Entities}(\mathcal{U}_t)}_{\text{supporting facts}}, \mathcal{H} \Big)$$

a_t: answer of sub-query t $\mathcal{U}_t$: its supporting atomic-fact hyperedges $\mathcal{H}$: all hyperedges

HyperProve at a glance

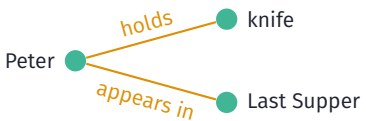


1 **Offline:** build hypergraph once 2 **Online:** loop over sub-queries 3 **Answer:** integrate / fallback

Offline: atomic facts as hyperedges

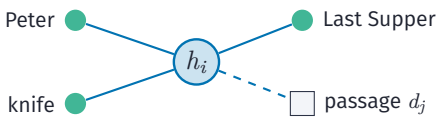
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Binary triples



holds the knife where? ⇒ context collapse

Atomic-fact hyperedge



x_i: *"In The Last Supper, Peter is holding a knife."*

- One fact = one hyperedge:
  $h_i = (x_i,\ \mathcal{A}_i,\ d_{\sigma(i)})$
  fact text, entities, source passage
  ⇒ keeps n**-ary context** & **provenance**

Hyperedge graph (fact ↔ fact):

$$W_s = \underbrace{B_{he} D_e^{-1} B_{he}^\top}_{\text{shared entity}} + \underbrace{B_{hd} D_d^{-1} B_{hd}^\top}_{\text{same passage}}$$

- W_s connects facts across passages via **shared entities** and within a passage via **shared context**, enabling multi-hop evidence chains.
- MuSiQue: 11.7K passages → **106K** hyperedges

Online ①: decompose, rewrite, and seed each hop

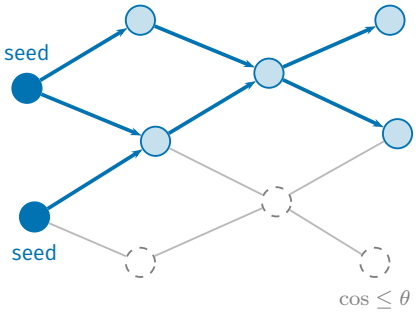


$$\mathcal{S}_t = \underbrace{\mathrm{TopK}(\mathcal{C}_t, K_c)}_{\text{carried seeds}} \cup \underbrace{\mathrm{DenseTopK}(q'_t, \mathcal{H}, K_d)}_{\text{fresh seeds}}$$

- **Decompose** $Q \to (q_1, \ldots, q_m)$ into dependent sub-queries
- **Rewrite** q'_t: fill references with validated answers
- **Carried seeds** (K_c=10): stay on the evidence chain
- **Fresh seeds** (K_d=10): escape a wrong neighborhood
- Unanswered hop $\to$ carry top-K_c partial hyperedges

↷ carry answer + supporting hyperedges into the next hop

Online ②: query-biased local hypergraph walk



$$T_s = \mathrm{RowNormalize}\big(W_s - \mathrm{diag}(W_s)\big),$$
$$T^\star_{s,t} = \mathrm{RowNormalize}\big(T_s|_{G^\star_t}\big).$$

Mix **structure** and **current sub-query**:

$$M_t = \alpha\, T^\star_{s,t} + (1-\alpha)\, T^\star_{q,t}$$

$$\boldsymbol{\pi}_t^{(r+1)} = (1-\gamma)\, u^\star_t + \gamma\, M_t^\top \boldsymbol{\pi}_t^{(r)}$$

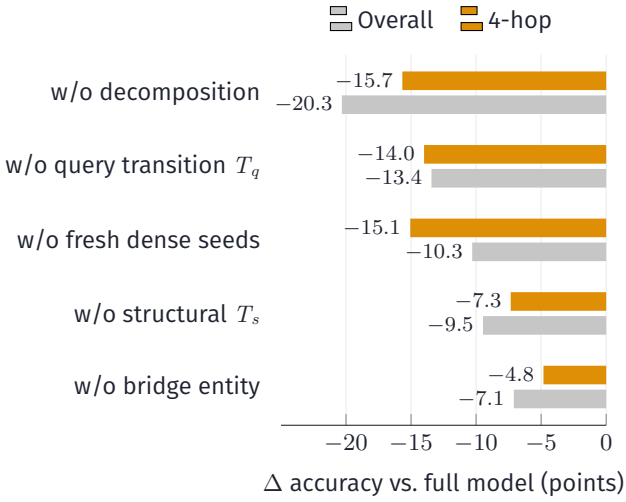
- $T^\star_{s,t}$: local structural transition on $G^\star_t$ (≤ 600 facts)
- $T^\star_{q,t}$: softmax over neighbors with $\cos(q'_t, x_k) > \theta$
- $u^\star_t$: restart at seeds $\mathcal{S}_t$ (personalized PageRank)
- Ranks **facts**; passages recovered via σ
- End: integrate answers or reader fallback

Main results: best on all three benchmarks

Family	Method	MuSiQue Acc	MuSiQue F1	HotpotQA Acc	HotpotQA F1	2Wiki Acc	2Wiki F1
Dense	BGE-M3	51.50	39.70	68.12	61.80	72.18	54.50
Structured	HippoRAG2	52.00	48.60	81.70	75.50	86.40	71.00
	PropRAG	50.90	46.73	85.80	73.79	77.50	68.78
	HyperGraphRAG	55.13	53.80	77.89	73.80	77.72	71.30
	HGRAG	58.60	55.04	80.10	69.43	85.00	74.10
Iterative	HopRAG	67.18	54.90	85.51	68.26	86.66	76.06
	ChainRAG	60.01	54.15	85.46	74.66	82.48	74.39
	HyperProve	**71.60**	**59.23**	**91.30**	**76.64**	**91.40**	**80.01**
	Δ vs. best	+4.42	+4.19	+5.50	+1.14	+4.74	+3.95

Best baseline *alternates* between structured and iterative families; HyperProve couples both.

Every component matters — most at 4 hops



- **Decomposition** is the foundation: -20.3 overall
- **Fresh seeds**: -6.8 on 2-hop but -15.1 on 4-hop — the carried frontier alone can trap retrieval
- **Query transition** matters more than structure: walking without q'_t expands into irrelevant facts
- **Bridge entity**: -7.1 — answers link hops that similarity misses

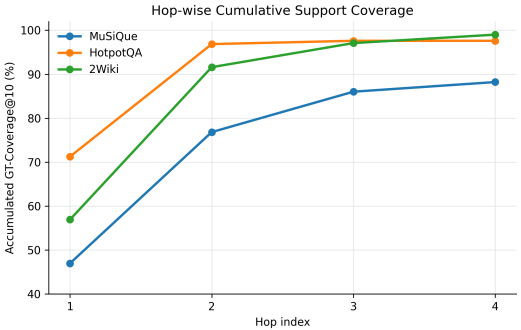
Full model: 71.6 overall, 49.4 on 4-hop MuSiQue (518 / 316 / 166 with 2 / 3 / 4 hops)

Better evidence chains, not just better answers

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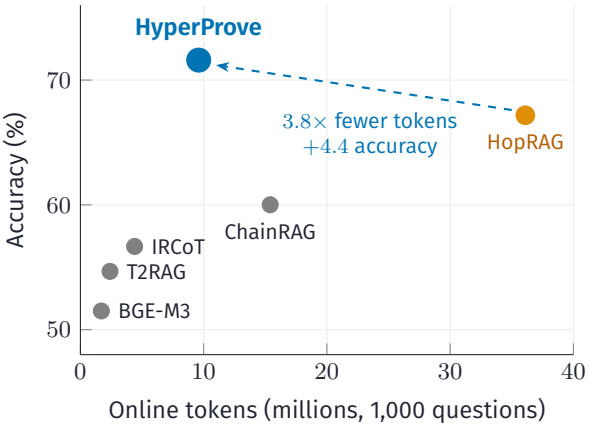
Method	Recall@10 MuSiQue	Hotpot	2Wiki	Gain after hop 1
HyperProve	**82.31**	**96.20**	**98.15**	**+41.24**
ChainRAG	67.42	89.45	75.85	+26.35
HopRAG	66.63	82.82	84.17	+25.98

- Same top-10 budget; fraction of gold passages retrieved
- +14.9 / +6.8 / +14.0 over the best iterative baseline
- Later hops recover **new** gold passages — not just longer context



HyperProve: cumulative gold coverage per hop

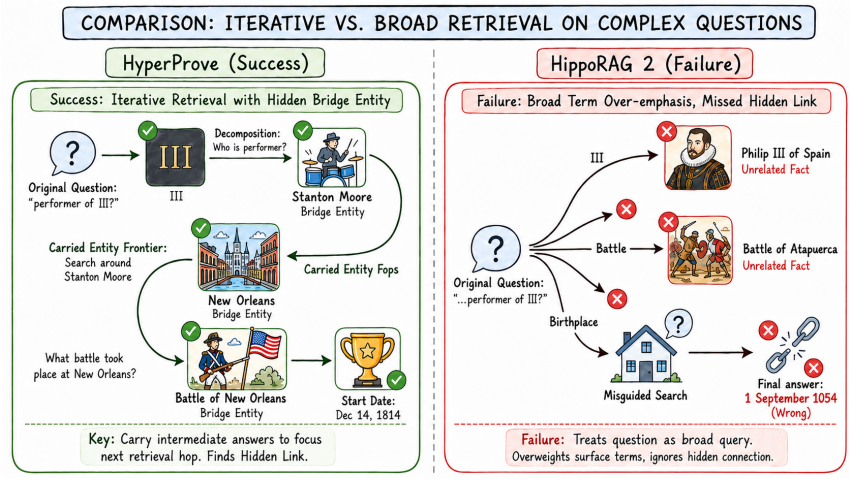
Better accuracy–cost trade-off



- vs. HopRAG: +4.42 acc, 26.5M fewer tokens, 13.3K s less latency
- vs. ChainRAG: +11.59 acc, 5.8M fewer tokens
- BGE-M3 / IRCoT / T2RAG are cheaper but trail by 15–20 points
- **Offline index**: 23.3K LLM calls, 20.8M tokens — on par with PropRAG (21.1M); one-time and parallel

MuSiQue; judge cost excluded.

Case study: following a hidden bridge



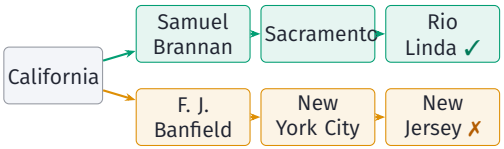
Takeaways and limitations

Takeaways

1. In multi-hop QA, relevance changes after every hop $\Rightarrow$ retrieval needs **state**
2. Carry the **answer** $+$ **its supporting atomic facts** as a structural frontier; walk locally, biased by the rewritten sub-query
3. Best Acc / EM / F1 on MuSiQue, HotpotQA, 2Wiki; $+14.9$ Recall@10 on MuSiQue; $3.8\times$ fewer tokens than HopRAG

Limitation: wrong bridges cascade

4-hop MuSiQue: gold chain vs. our prediction



- Unsupported bridge steers *both* rewrite and frontier $\rightarrow$ confidence-aware or multi-frontier propagation
- LLM cost offline $+$ online $\rightarrow$ distillation
- LLM-judge, static benchmarks

References

[1] P. Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS* 2020.

[2] B. J. Gutiérrez et al. From RAG to Memory: Non-Parametric Continual Learning for LLMs (HippoRAG2). *ICML* 2025.

[3] J. Wang, J. Han. PropRAG: Guiding Retrieval with Beam Search over Proposition Paths. *EMNLP* 2025.

[4] H. Luo et al. HyperGraphRAG: RAG via Hypergraph-Structured Knowledge Representation. *NeurIPS* 2025.

[5] C. Wang et al. Cross-Granularity Hypergraph RAG for Multi-hop Question Answering. *AAAI* 2026.

[6] H. Trivedi et al. Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. *ACL* 2023.

[7] H. Liu et al. HopRAG: Multi-Hop Reasoning for Logic-Aware Retrieval-Augmented Generation. *Findings of ACL* 2025.

[8] R. Zhu et al. Mitigating Lost-in-Retrieval Problems in Retrieval Augmented Multi-Hop QA. *ACL* 2025.

[9] H. Trivedi et al. MuSiQue: Multihop Questions via Single-hop Question Composition. *TACL* 2022.

Thank you!

Carry the answer **and** its evidence.

HyperProve: Answer-Guided Hypergraph Expansion for Multi-Hop Question Answering

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Backup Slides

Backup: full results (Acc / EM / F1)

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Method	MuSiQue			HotpotQA			2Wiki		
	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1
Contriever	47.80	22.30	32.10	55.55	44.40	57.50	69.60	38.50	43.70
BGE-M3	51.50	27.80	39.70	68.12	47.80	61.80	72.18	48.20	54.50
HippoRAG2	52.00	37.20	48.60	81.70	62.70	75.50	86.40	65.00	71.00
PropRAG	50.90	29.30	46.73	85.80	57.80	73.79	77.50	58.00	68.78
HyperGraphRAG	55.13	42.20	53.80	77.89	63.90	73.80	77.72	68.30	71.30
HGRAG	58.60	41.19	55.04	80.10	54.90	69.43	85.00	64.10	74.10
IRCoT	56.67	34.10	47.60	85.23	55.70	71.20	83.65	60.70	74.30
T2RAG	54.67	34.30	45.60	81.33	54.20	67.30	83.75	66.70	74.40
HopRAG	67.18	42.40	54.90	85.51	61.10	68.26	86.66	62.00	76.06
ChainRAG	60.01	41.20	54.15	85.46	60.70	74.66	82.48	63.30	74.39
HyperProve	**71.60**	**42.83**	**59.23**	**91.30**	**63.92**	**76.64**	**91.40**	**69.70**	**80.01**

Std over 3 runs ≤ 0.80. EM margins on MuSiQue (42.83 vs. 42.40) and HotpotQA (63.92 vs. 63.90) are within run-level variation. Back

Backup: hypergraph and transition details

Incidences and structural transition (N hyperedges, M_e entities, M_d passages)

$$B_{he}[i,a] = \mathbf{1}[e_a \in \mathcal{A}_i^+], \quad B_{hd}[i,j] = \mathbf{1}[\sigma(i) = j], \quad \mathcal{A}_i^+ = \mathcal{A}_i \cup \mathrm{Syn}(\mathcal{A}_i)$$

$$T_s = \mathrm{RowNormalize}\big(W_s - \mathrm{diag}(W_s)\big), \quad W_s = B_{he} D_e^{-1} B_{he}^\top + B_{hd} D_d^{-1} B_{hd}^\top$$

Query-biased transition on local subgraph $G_t^\star$, with $c_{t,k} = \cos\big(\phi(q_t'), \phi(x_k)\big)$:

$$T_{q,t}^\star[i,k] = \frac{\exp(c_{t,k}/\tau)}{\sum_{\ell \in \mathcal{N}_{t,\theta}(i)} \exp(c_{t,\ell}/\tau)} \text{ if } k \in \mathcal{N}_{t,\theta}(i) = \{k \in \mathcal{N}_t(i) : c_{t,k} > \theta\}, \text{ else } 0$$

- Empty gated row $\Rightarrow$ walk falls back to the structural transition
- $\tau = 1,\ \theta = 0.4,\ \alpha = 0.5,\ \gamma = 0.85,\ \eta = 0.8$; local subgraph ≤ 600 hyperedges
- Reader fallback: $s(d_j) = \max_{t,\, h_i \in \mathcal{H}_t,\, \sigma(i)=j} \pi_t(h_i)$, top-10 passages + ranked facts

Backup: HyperProve algorithm (condensed)

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$(q_1, \ldots, q_m) \leftarrow$ DECOMPOSE(Q); $\mathcal{C}_1, \mathcal{R} \leftarrow \emptyset$
for $t = 1$ to m **do**
  $q'_t \leftarrow$ REWRITE$(q_t, \mathcal{R})$; $\mathcal{S}_t \leftarrow$ TOPK$(\mathcal{C}_t, K_c) \cup$ DENSETOPK$(q'_t, \mathcal{H}, K_d)$
  $(\mathcal{H}_t, \boldsymbol{\pi}_t, \mathcal{P}_t) \leftarrow$ LOCALWALK$(\mathcal{S}_t, q'_t)$ ▷ $M_t = \alpha\, T^{\star}_s + (1-\alpha)\, T^{\star}_q$
  $(s_t, a_t, \mathcal{U}_t) \leftarrow$ EVALUATE$(q_t, \mathcal{H}_t, \mathcal{P}_t)$
  if $s_t =$ ANSWERED **then**
    $\mathcal{R} \leftarrow \mathcal{R} \cup \{(q_t, a_t)\}$; $\mathcal{C}_{t+1} \leftarrow$ FINDHYPEREDGES$(\{b_t\} \cup$ ENTITIES$(\mathcal{U}_t))$
  else $\mathcal{C}_{t+1} \leftarrow$ TOPK$(\mathcal{H}_t, \boldsymbol{\pi}_t, K_c)$
  end if
end for
return all answered ? INTEGRATEANSWERS$(Q, \mathcal{R})$: READERFALLBACK$(Q, \mathcal{P}_{10}, \mathcal{M}_H)$

Backup: sensitivity and reader fallback (MuSiQue)

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Sensitivity to α and θ

α	θ	Recall	EM	F1	Acc
0.50	0.40	82.31	42.83	59.23	71.60
0.25	0.40	81.18	42.41	58.76	70.80
0.75	0.40	83.06	43.12	59.61	71.40
0.50	0.30	83.45	41.97	58.49	70.30
0.50	0.50	81.72	43.05	59.40	71.90

Accuracy varies by ≤ 1.3 points; default kept (no post-hoc tuning).

Reader-fallback frequency

Depth	Fallbacks	Questions	Rate
2-hop	41	518	7.9%
3-hop	38	316	12.0%
4-hop	44	166	26.5%
Overall	123	1,000	12.3%

Deeper questions leave more room for an unresolved hop.

Backup: offline cost and dataset statistics

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Offline construction (MuSiQue, 11,656 passages)

Method	Extraction	Calls	Tokens	Cost source
HyperGraphRAG	facts + entities	11,656	20.9M	normalized estimate
PropRAG	entities + propositions	23,312	21.1M	reported
HGRAG	entities	11,656	5.7M	normalized estimate
HyperProve	entities + facts	23,312	20.8M	measured

Evaluation subsets (test split, following PropRAG)

Dataset	Questions	Passages	Entities	Hyperedges
2WikiMultiHopQA	1,000	6,119	57,906	55,032
HotpotQA	1,000	9,812	112,308	99,394
MuSiQue	1,000	11,656	117,894	105,542